Submission Summary

Conference Name	International Conference on Electrical, Communication, and Computing Technologies, IInd Edition
Track Name	Track 3: Advanced Computing and Secure Systems
Paper ID	516
Paper Title	A CONFIGURABLE PERFORMANCE MONITORING AND ANALYTICS IP FOR RISC-V PROCESSORS
Abstract	Modern RISC-V processor Performance Monitoring Units (PMUs) measure a range of events such as branches, cache operations, executed instructions and cycles, and display raw number counts. These metrics need to be manually analyzed by engineers through scripts or a waveform viewer to provide answers to simple questions such as "Why is the IPC low?". This soft IP is a stealthy observer in the RISC-V pipeline, gathering raw information about events in the hardware and making it available as rich performance data without changing the basic design of the hardware. Microarchitectural Analysis, Architectural Profiling are two modes of the IP. There are 5 re-usable components of the IP: Event Collector, Event Counters, Analytics Engine, Mode Controller and Register Bank. An example is provided of how the IP can generate derived metrics without any human post processing in the hardware counters. Can support different RISC-V implementations with analytics. The proposed Soft IP was tested with the benchmark

	workloads with 10K and 2K instructions and automatically collected raw PMU events and derived performance metrics like IPC (0.92), CPI (1.07–1.08), 92% pipeline efficiency, 92% stage utilization etc providing these insights instantaneously and drastically reducing the dependency on manual post-processing and analysis time. In addition, it explicitly studied the memory operation ratio, hazard distribution and forwarding efficiency (100%) of the hardware. The results show that the IP can be easily integrated into a variety of RISC-V processor architectures, and the IP is able to provide accurate architectural and microarchitectural performance information.
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